TM Forum Introductory Guide

Shaping the Future Internet of Agents: Agentic Collaboration through Semantics and Value

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Executive Summary

The aim of this document is to catalyse a foundational evolution in how digital ecosystems operate and create value. A visionary framework of agentic collaboration in digital ecosystem is proposed to achieve intelligent, autonomous, and sustainable collaboration across CSPs and other digital service providers from telecom and other vertical industries by applying agentic AI systems with ontology-driven semantics interoperability.

This document introduces the concept of the Internet of Agents (IoA), an open digital stratum where autonomous, AI-powered agents/systems interact to coordinate tasks, share resources, and execute transactions with minimal human intervention. It highlights the shift from connecting passive objects to enabling a society of autonomous digital actors and explores the potential of IoA as a dynamic, self-organizing economic ecosystem where value is created, traded, and monetized among autonomous participants.

Target Audience of these documents includes ecosystem strategists, enterprise architects, CTO/CIO offices, product managers, standards participants, and solution providers in telecommunications and adjacent High-tech industries.

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Introduction

The telecommunications and high-tech industries are undergoing a fundamental structural shift, moving beyond traditional, linear value chains towards dynamic, multi-party digital ecosystems. This shift from linear value chains to interconnected networks of partners, suppliers, and customers is driven by the demand for innovative, composite services, such as networked AI, adaptive edge computing, and integrated enterprise solutions. Success in this landscape is no longer solely about internal optimization but increasingly about the ability to seamlessly and intelligently collaborate across organizational and technological boundaries.

However, the current reality is hampered by significant architectural and operational fragmentation. These silos, coupled with proprietary data models, lead to chronic challenges: painfully slow integration cycles, brittle point-to-point interfaces, and an inherent inability to dynamically discover, negotiate, and compose services in real-time. To break these silos and achieve fluid ecosystem collaboration, a foundational shift is required. The solution lies in establishing a **semantically precise, machine-readable "common language" for the entire ecosystem**.

Shared ontologies move beyond simple data dictionaries to define concepts, their properties, and their relationships within a logical, unambiguous way that both humans and machines can interpret consistently. When all participants—systems and APIs—adhere to a shared conceptual model, semantic interoperability is achieved, data becomes inherently contextualized, and misunderstandings are eliminated. Then, the second challenge is how semantically precise, machine-readable ontologies provide a "common language," and how agentic AI systems can use this language to autonomously negotiate, discover, and collaborate to fulfil complex objectives.

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1. Project Scope and Objectives

1.1. Scope

This document presents a visionary framework for the next generation of digital ecosystems. It introduces the concept of the **Internet of Agents (IoA)**, a decentralized digital stratum where autonomous, AI-powered agents collaborate, negotiate, and transact with minimal human intervention. The scope is deliberately toward a more autonomous and intelligent ecosystem by defining a new approach, centered on ontology-enabled agentic collaboration.

The document aims to provide a strategic and architectural guide for Communication Service Providers (CSPs), High-Tech companies, and broader ecosystem participants. It focuses on:

Business Architecture Alignment: Incorporating the concepts of value networks and capability frameworks to ensure technical models are grounded in business objectives, enabling the dynamic composition and monetization of capabilities across organizational boundaries.

The Architectural Shift: Defining the components and interaction models for a collaboration framework where autonomous software agents, powered by shared ontologies, act on behalf of entities within a value network.

The Semantic Foundation: Establishing the critical role of formal, shared ontologies (aligned with and extending industry standards like the TM Forum SID) as the foundational layer for unambiguous understanding, enabling true semantic interoperability beyond syntactic API alignment.

Cross-Domain, Cross-Vendor Context: Addressing collaboration challenges and solutions that span different administrative domains (e.g., between a CSP, an enterprise, and a cloud provider) and different vendor technology stacks.

Evolving Industry Frameworks: Exploring a pathway for evolving current standards and architectures to potentially incorporate AI and data as a semantic and agentic layer. The document examines how ODA could be enhanced to serve as a trusted environment for agent discovery, negotiation, and transaction, thereby making the vision of a dynamic, self-optimizing digital ecosystem an achievable reality.

Integration with Ecosystem Frameworks: Explicitly mapping the proposed agentic approach to the TM Forum's foundational work, including the ODA, the Information Framework (SID), the Capability Model, the Ecosystem Reference Architecture (GB1067), and the Zero-Touch Partnering framework (GB1027).

The document scopes a framework, principles, and a forward-looking exploration—not a specification of finalized solutions. It is intended to catalyse discussion and guide future development toward an **Internet of Agents (IoA)**.

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2. Vision

The transition with this agentic collaboration in digital ecosystem is a fundamental realignment to unlock new business value and operational agility. The ecosystem is no longer a static map of fixed connections but a proactive, self-optimizing marketplace of capabilities. Here, software systems/agents, representing the capabilities and intents of various participants, can dynamically discover each other, negotiate and collaborate tasks, form transient agreements, coordinate actions to fulfil complex business objectives and co-create value. In this agentic ecosystem, the agents collectively form an Internet of Agents (IoA) creating a fluid and responsive value network that continuously adapts to opportunity and demand with unprecedented speed, agility, and precision.

Figure 1: The Evolving Agentic Ecosystem

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This vision builds upon the foundations laid by the Internet of Things (IoT) and the proliferation of web services. However, the key difference lies in the shift from connecting passive objects or facilitating human-centric services to enabling a society of autonomous digital actors. These agents are not merely conduits for data or tools for human interaction; they are active participants in the digital economy, capable of independent decision-making and action.

In this vision: An enterprise's demand for a mission-critical, low-latency application triggers a real-time auction and composition of resources—from connectivity and edge compute to security functions—across multiple providers, orchestrated not by a central planner but by the interacting agents themselves. The entire value network can dynamically reconfigure in response to disruptions, market opportunities, or new intents, fostering an environment of continuous innovation and resilience.

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3. Background

The telecommunications and high-tech industries are engaged in a fundamental structural transformation, evolving from traditional, vertically integrated models into dynamic and interconnected digital ecosystems. This shift from linear value chains to multi-party networks is driven by the market demand for innovative, composite services—such as networked AI, adaptive edge computing, and integrated enterprise solutions—that no single entity can deliver alone. Success in this new landscape is contingent upon the ability to seamlessly and intelligently collaborate across organizational and technological boundaries.

However, the industry's current architectural foundation is inadequate for this. The ecosystem integration and collaboration is primarily enabled by point-to-point APIs and static orchestration. This results in chronic operational challenges:

Brittle and Slow Integration: Each new partnership requires extensive, custom integration projects, leading to months-long cycles, high costs, and a fragile mesh of connections that stifles agility.

Static and Non-Adaptive Interactions: Pre-defined workflows and contracts cannot dynamically respond to real-time changes in context, demand, or capability, locking partners into rigid, inefficient engagements.

Semantic Misalignment: Even with standardized APIs, the underlying meaning of data and services is often ambiguous. Without a shared understanding of concepts like "service availability," "quality," or "cost," automated processes break down, requiring continuous human intervention for reconciliation.

The core impediment is two-fold: a lack of **semantic interoperability** and a lack of **collaborative agency**. First, while standardized APIs exist, they lack a shared, machine-readable understanding of the data they convey. This semantic gap forces extensive manual mapping and constant maintenance, making every integration a custom project. Second, even with perfect data understanding, current systems lack the autonomous "agency" to proactively engage in complex, multistep collaborations. Processes remain static, pre-programmed, and dependent on human intervention for any deviation or new opportunity.

These operational fragmentations directly inhibit strategic progress. It is the primary barrier to achieving higher levels of Autonomous Networks (AN) at Levels 4 and 5, which require intent-driven, self-optimizing systems capable of operating across ecosystem borders. It stifles the vitality of the ecosystem by making the onboarding of innovative niche players prohibitively difficult, and it threatens sustainability by locking capital and operational expenditure into maintaining brittle, non-scalable integration fabrics.

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4. Internet of Agents (IoA) and Agentic Ecosystem

4.1. Internet of Agents

The Internet of Agents (IoA) is a decentralized and open digital stratum in which autonomous, goal-driven agents/agentic systems (empowered by artificial intelligence) - acting on behalf of humans, enterprises, or machines - discover, communicate, negotiate, collaborate, and transact with one another across heterogeneous environments to dynamically coordinate tasks, share resources, and execute complex transactions with minimal human intervention.

The Internet of Agents (IoA) represents a paradigm shift in how we interact with technology and conduct business in the digital realm. This paradigm shift is from connecting passive objects or facilitating human-centric services to enabling a society of autonomous digital actors that can act on behalf of users, systems, or organizations. Building on top of the foundational architecture, the true disruptive potential of the Internet of Agents (IoA) lies in its emergence as a dynamic, self-organizing economic ecosystem. This perspective moves beyond technical components to examine how value is created, traded, and monetized among autonomous participants.

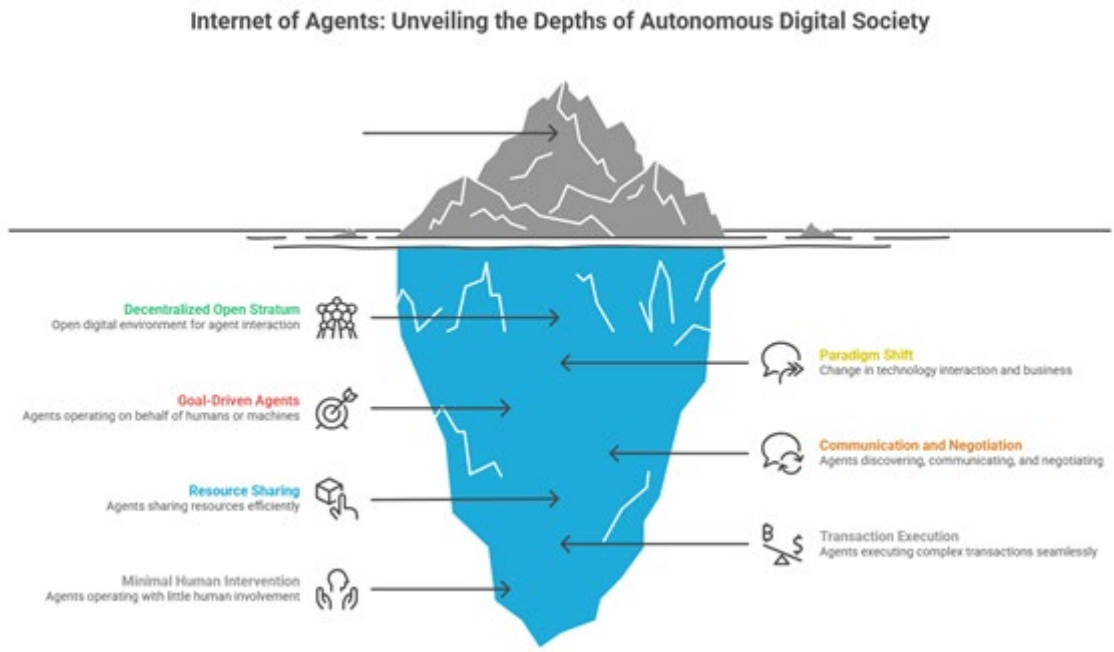

Figure 2: Internet of Agents: Unveiling the Depths of Autonomous Digital Society

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- The agents within the IoA are characterized by their autonomy, goal-driven nature, and intelligence. Autonomy allows them to operate independently, without constant human supervision. Their goal-driven nature ensures that they are focused on achieving specific objectives, whether it's optimizing energy consumption, managing supply chains, or providing personalized services. Intelligence, powered by artificial intelligence (AI) and machine learning (ML), enables them to learn from experience, adapt to changing circumstances, and make informed decisions.
- The architecture of the IoA is envisioned as a decentralized and open digital stratum. Decentralization is crucial for ensuring resilience, security, and fairness. By distributing control and data across a network of nodes, the IoA avoids the single points of failure and centralized control that can plague traditional systems. Openness is equally important for fostering innovation and interoperability. By adhering to open standards and protocols, the IoA allows different agents and systems to seamlessly interact with one another, regardless of their underlying technology or origin.
- The potential applications of the IoA are vast and span across various industries and domains. In supply chain management, agents can autonomously track goods, optimize logistics, and negotiate contracts with suppliers. In energy management, they can balance supply and demand, optimize energy consumption, and trade energy on the grid. In healthcare, they can monitor patients' health, schedule appointments, and provide personalized treatment plans. In finance, they can execute trades, manage investments, and detect fraud.

4.2. Agentic Collaboration

Agentic collaboration is the core operational principle and value-creation mechanism for the Internet of Agents (IoA). Agentic collaboration evolves beyond pre-programmed, static interaction flows into a dynamic, adaptive, and goal-driven synergistic model. Its essence lies in multiple autonomous agents/systems (representing enterprises, individuals, or machines) achieving complex objectives through dynamically form temporary alliances, negotiate terms, coordinate task, and share resources to fulfil complex objectives. This transcends simple service invocation to a structured yet fluid process of joint problem-solving and execution among digital entities with their own agency.

The collaborative dynamics within an Internet of Agents (IoA) ecosystem move beyond static, pre-defined workflows to enable fluid, intelligent, and goal-driven interactions.

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Dimension	Current Systems and Business Interactions	Agentic Collaboration in IoA
Nature of Interaction	**Static, Passive, Point-to-Point**: Based on pre-defined APIs and contracts, interaction flows are fixed, and changes are costly.	**Dynamic, Active, Networked**: Agents proactively discover and initiate collaboration; multi-actor networks formed in real time; interaction patterns are generated dynamically based on task requirements.
Coordination Mechanism	**Pre-Orchestrated:** Workflows and business logic are centrally designed and hard-coded.	**Emergent & Negotiated:** Coordination plans emerge from agent interaction, negotiation, and local decision-making.
Temporal Scope	**Long-Term Contracts:** Relationships and SLAs are fixed for months or years.	**Moment-to-Moment Agreements:** Alliances and service terms are established for the duration of a specific task or transaction.
Value Creation	**Linear, Pre-defined**: Value is transferred along a fixed value chain; innovation is limited by existing partners and integration architecture.	**Emergent, Co-created**: Value is co-created within dynamically formed networks; new value propositions can be rapidly realized through previously unimaginable agent combinations .
Semantic Understanding	**Syntactic Interoperability**: Even with standardized APIs, the **semantics** (meaning) of data still rely on manual interpretation and mapping, prone to ambiguity.	**Semantic Interoperability**: Through shared ontologies, agents possess a **machine-understandable common semantics** for data, services, and goals, laying the foundation for automated collaboration .
Level of Autonomy	**Instrumental, Requires Human Intervention**: Systems operate strictly on instructions; exception handling and complex decisions heavily depend on humans.	**Goal-Driven, Highly Autonomous**: Agents are given high-level goals and can autonomously plan, decide, execute, and reflect based on feedback, with humans playing more of a role as goal-setters and supervisors .

Agentic collaboration ranges from fleeting, transactional teams to persistent, co-optimizing networks, incorporating hierarchical delegation, market-like negotiations, and collective reasoning to create adaptive, intelligent, and efficient multi-agent systems. Some typical scenarios and characteristics including:

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Dynamic On-Demand Teaming / Transient Task-Specific Alliances: This is the foundational collaborative form, where agents dynamically discover each other and form temporary, purpose-built teams to achieve a specific, often complex objective (e.g., delivering an immersive AR experience or deploying a mission-critical application). The alliance is strictly transient, dissolving automatically upon task completion or when conditions change. This model offers extreme flexibility, contrasting sharply with today's long-term, statically contracted partnerships and rigid technical integrations.

Hierarchical Task Execution and Cascading Delegation: For complex goals, a coordinator agent can autonomously decompose the objective into sub-tasks and delegate them to specialized agents. These sub-agents may further delegate, creating a dynamic, multi-level hierarchy formed on-the-fly based on capability discovery and trust metrics. This resembles an automated "general contractor-subcontractor" model, enabling adaptive execution that is more resilient to uncertainty than pre-defined, rigid workflow processes (like BPMN).

Competitive-Cooperative Negotiation and Consensus-Building: Collaboration in IoA is not always purely cooperative; it often involves competitive or deliberative elements to optimize outcomes.

- Negotiation Swarms: For resource allocation, agents representing demand (e.g., for network slices) interact with swarms of supply agents from multiple providers. Collaboration occurs through real-time auctions and multi-variable negotiations (price, SLA, carbon footprint), dynamically composing the best-fit consortium.
- Consensus-Building Debates: For open-ended problems or decisions, agents may propose competing solutions or viewpoints based on different data or expertise. They engage in structured debates or evaluation mechanisms to reach a superior collective decision or present vetted options to a user, introducing diversity and critical thinking rarely found in current systems.

Collective Intelligence and Co-Creation: Agents collaborate not merely to share resources but to synthesize novel information, models, or strategies. By integrating diverse capabilities and data sources (e.g., from logistics, weather, and traffic systems), they can co-create adaptive solutions—like a dynamic routing model—that exhibit collective intelligence exceeding any single system's capability. This extends beyond one-off task completion to continuous, synthetic problem-solving.

Continuous Co-Optimization: In persistent ecosystems like supply chains or power grids, agent collaboration is an ongoing process. Agents representing different nodes of a value network continuously share status and sensor data (inventory, delays, demand shifts). They adjust their behaviours in real-time not just for local optimization, but to jointly respond to disruptions and seize opportunities, thereby achieving system-wide adaptation and resilience through perpetual, collaborative adjustment.

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To realize this vision of agentic collaboration, the digital ecosystem should evolve to meet new foundational requirements, enabled by a convergence of advanced technologies:

Semantic Interoperability & Shared Ontologies: A machine-readable, shared understanding of everything—services, capabilities, quality metrics, commercial terms, and operational states. This is the antidote to the "semantic misalignment" that plagues current APIs.

- Domain Ontologies and Knowledge Graphs. Agents should be able to use formal, logic-based ontologies (e.g., based on OWL, RDF) to describe their capabilities and needs. A shared "ecosystem lexicon" for concepts like ServiceLatency, CarbonCredit, or DataSovereigntyRule is essential for unambiguous automated negotiation and collaboration.

Agent Autonomy with Governed Agency: Agents will need the authority to make decisions (e.g., select a partner, agree on a price) within a clearly defined scope of action and a set of governance rules.

- AI/ML for Decision-Making & Strategy: Reinforcement learning for negotiation strategies, federated learning for collaborative model improvement, and causal reasoning for understanding the impact of actions.
- Policy Engines & Smart Contracts:Codified rules (e.g., "never collaborate with agents below this trust score") and executable contracts on distributed ledger technology (DLT) provide the governance framework for autonomous action.

Dynamic Discovery & Trust Fabric: Agents cannot collaborate if they cannot find each other. They need to have a basis for trust, assessing the reliability, performance history, and credibility of potential partners.

- Decentralized Identity & Verifiable Credentials (DID/VCs):Agents have persistent, self-sovereign identities and carry machine-verifiable attestations of their capabilities or compliance.
- Reputation Systems:Decentralized ledgers or protocols that track and provide access to an agent's collaborative history and performance metrics.

Interaction Protocols & Conversation Frameworks: Standardized "languages of collaboration" beyond simple API calls. These protocols define the possible conversation flows for discovery, negotiation, commitment, and collaboration.

- Interaction Protocol Standards:Formal specifications for auctions, contract-net protocols, and coalition formation. These provide the "rules of engagement" for agentic discourse.

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Collective Learning & Ecosystem Intelligence: The ecosystem as a whole should learn and improve from the myriad interactions occurring within it, identifying systemic optimizations and novel value-creation patterns.

- Privacy-Preserving Analytics & Federated Learning:Allows agents to contribute to and benefit from collective learning without exposing proprietary data.
- Mechanism Design & Ecosystem Simulation:Advanced modeling to design incentive and interaction rules that lead to desired, stable, and efficient ecosystem-wide outcomes.

This shift is not merely technical, it is the enabler of new business models and value creation. It allows for the dynamic formation of complex, co-created services, real-time micro-transactions across partners, and resilient value networks that can automatically adapt to disruption. By embedding intelligence and semantics into the fabric of collaboration, the ecosystem gains the dynamic vitality to rapidly innovate and the structural sustainability to evolve efficiently.

4.3. Agentic Ecosystem

With all the foundations to build IoA infrastructure and applications, the true disruptive potential of the IoA lies in its emergence as a dynamic, self-organizing economic ecosystem. This perspective moves beyond the technical components of the IoA to examine how value is created, traded, and monetized among autonomous participants. In this ecosystem, agents can act as both consumers and producers of goods and services, engaging in complex transactions and collaborations to achieve their goals.

The IoA ultimately points toward the emergence of an "Autonomous Economy"—a vast network of AI-driven entities trading specialized skills and digital resources with minimal human intervention. The battleground for platform dominance will shift from user attention (the web) or physical logistics (e-commerce) to orchestrating trust and value flow in a machine-to-machine marketplace. The companies that will thrive are those that start architecting their services as composable, billable agent units today, while actively participating in the development of the open economic protocols that will underpin this new frontier.

The monetization of agent-driven services requires to be automated, granular, and trust-minimized economic mechanisms. Traditional invoicing is impossible; the system needs machine-native payment protocols.

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Mechanism	How It Works	Applications
Micropayment & Streaming Payments	Tiny, real-time payments for continuous service use (e.g., pay-per-millisecond-of-API-call, per-data-row-processed). Enabled by cryptographic payment channels or central platform credits.	An analytics agent streams processed sensor data to a monitoring agent. Payment flows concurrently as a "data stream," stopping instantly if service degrades.
Smart Contract-Based Revenue Sharing	Pre-programmed, immutable contracts on a ledger automatically split revenue upon verification of task completion. Rules (splits, conditions) are transparent and self-executing.	A travel planning agent sells a trip package. Upon completion, the smart contract automatically splits payment: 60% to the airline-booking agent, 25% to the hotel agent, 10% to the local experience agent, 5% to the orchestrator.
Bidding & Auction Models	Agents participate in real-time markets (spot auctions, reverse auctions) to bid for tasks or offer services, establishing dynamic market prices.	A logistics company's agent needs to fill last-minute cargo space. It publishes the slot; fuel-optimizer agents and routing agents bid to provide savings, with the winning bid executing automatically.
Staking & Reputation Bonding	Agents or their owners should be able to stake value (eg. tokens) as a collateral bond to participate in high-value tasks. Penalties for SLA violations (slashing) are automated, ensuring trust.	To be listed in a high-reliability manufacturing diagnostics marketplace, an agent should be able to stake certain conditions, guaranteeing its performance and insulating buyers from failure costs.
Subscription & Capacity Tokens	Human users or enterprise agents purchase tokenized access passes (e.g., "10,000 inference credits," "24/7 monitoring for Q3") that are consumed as services are used.	An enterprise buys a "Supply Chain Resilience Bundle" of tokens, consumed by risk-assessment agents, alternative supplier-finder agents, and logistics re-router agents as needed throughout the quarter.

4.4. From Patterns to Internet

The explosive growth of the IoA will be fuelled not by isolated intelligent agents, but by sophisticated, autonomous interaction patterns between them. These patterns transform simple automation into dynamic, emergent economic activity.

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Interaction Pattern	Economic Mechanism	Value Created & Growth Driver
Auction & Dynamic Pricing	Agents act as buyers/sellers in real-time computational markets for resources (CPU, GPU, data), services, or physical goods.	Optimizes resource allocation at micro-scale. Drives growth by creating liquid markets for previously illiquid or fixed-capacity assets (e.g., unused network slices, idle warehouse robots).
Contract Net & Bidding	An "orchestrator" agent broadcasts a task; specialist "contractor" agents bid with capability and price. The most economically efficient bid wins.	Maximizes efficiency and specialization. Fuels a long-tail marketplace for hyper-specialized micro-services, allowing niche AI capabilities to find economic viability.
Consensus & Collective Decision	Multiple agents (e.g., representing supply chain partners) negotiate to reach consensus on a shared plan (e.g., production schedule) using voting or Byzantine fault-tolerant protocols.	Enables complex, multi-stakeholder coordination without centralized control. Reduces friction in forming dynamic consortiums for projects, accelerating collaborative ventures.
Delegate & Chain-of-Agents	An agent delegates a sub-task to another, creating a chain of value-add. Payments flow upstream, with each agent taking a margin for its role (coordination, specialization, access).	Creates layered value chains and revenue-sharing models. Encourages the professionalization of agents into specific roles (broker, auditor, insurer) within workflows.
Competitive & Adversarial	Agents compete to achieve opposing goals (e.g., a trading agent vs. a market-maker agent) within governed environments, leading to sophisticated strategic equilibria.	Drives innovation and resilience. Creates demand for advanced simulation environments and regulatory agents, a new sector in itself.

These patterns shift IT spending from static licensing (CAPEX/OPEX) to dynamic, outcome-based consumption. Economic activity explodes as transactions move from monthly invoices between companies to millions of micro-transactions between agents within seconds, creating a high-velocity, high-liquidity digital economy.

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For an agent from one ecosystem (e.g., a logistics optimizer) to reliably delegate a task via a Delegate & Chain-of-Agents pattern to an agent from another (e.g., an autonomous drone fleet manager), they should share a precise, unambiguous understanding of concepts like *DeliveryRoute*, *PackageIntegrity*, *WeatherConstraint*, and *ServiceLevelAgreement*. Without this, delegation becomes a game of misinterpretation, leading to failed transactions, economic disputes, and systemic fragility. This is where ontology-based collaboration becomes critical. The shared, formal ontologies of the Semantic Layer provide this common world model. They define the properties, relationships, and logical rules governing the entities within a domain.

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5. Evolving the Open Digital Architecture (ODA) to Enable the IoA

The TM Forum's Open Digital Architecture (ODA) offers a widely recognized blueprint for a composable, cloud-native enterprise. To explore its potential in supporting the requirements of an Internet of Agents (IoA), ODA may benefit from further evolution—from a framework primarily designed for B2B integration into a platform capable of dynamic orchestration for an agent-based economy. This potential evolution could be viewed as a strategic enhancement across ODA layers, exploring how it might serve as a standardized environment where agents can potentially discover, negotiate, and transact with various capabilities.

5.1. Embedding Semantics: From Syntactic APIs to Ontology-Driven Understanding

To support semantic interoperability, ODA's information and API layers could be extended with machine-readable meaning.

- **Information Systems Architecture - Ontology-Driven Semantics:** The Information Framework (SID) might be formalized into a machine-reasoning ontology (e.g., using OWL/RDF). This could help move the SID from a data dictionary into the shared conceptual model - the common world model - that agents can use to interpret domain concepts.
- **Open APIs - Semantic Enrichment:** TM Forum Open APIs could be extended with semantic annotations to describe their non-functional properties (cost, latency, geo-restrictions), negotiation endpoints, and dynamic SLA terms, etc. By doing so, the agents will be able to interpret autonomously via protocols such as A2A/MCP with guaranteed performance level.

5.2. Enabling a Dynamic Marketplace: From Static Registry to Agent Service Ecosystem

To facilitate dynamic discovery, ODA's component registry concepts could evolve toward as real-time, federated marketplaces.

- **Deployment & Runtime - Federated Service Registry:** The ODA Canvas could host a reference implementation for a decentralized, federated registry of agent-services, potentially allowing agents from different domains to discover each other by their capabilities.

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- **Deployment & Runtime - Advanced Discovery Mechanisms:** The registry could be paired with matchmaking engines - for example, to help identify combinations of agents that may fulfil complex, multistep goal based on capability, availability, trust, pricing or other parameters.
- **ODA Components - Agent Metadata Schema:** The component definition could be extended to include rich agent metadata, such as formal capability declarations (informed by ontologies), verifiable trust indicators, and references to associated AI models.

5.3. Building a Trust Fabric: Identity, Reputation, and Governance

To address trust and governance considerations, ODA's security and governance layers might incorporate more dynamic, runtime-aware elements.

- **ODA Governance - Runtime Agentic Control:** Governance approaches could expand from project-level oversight to include runtime mechanisms - for instance, an "Agentic Control Plane" to enforce dynamic policies-as-code that enforce certain constraints, real-time audit trails of agent decisions, or management of agent identities and verifiable credentials.
- **Deployment & Runtime - Reputation System:** The ODA Canvas could integrate with or provide a decentralized ledger for tracking and querying agent performance metrics, contributing to a verifiable reputation fabric that agents can use to assess the trustworthiness of potential collaborators when assessing potential collaborators.

5.4. Enabling Agent-Native Collaboration: From API Calls to Conversations

To support agentic collaboration, ODA's interaction model could be extended to accommodate multistep dialogues.

- **Open APIs & Components - Standardized Conversation Protocols:** ODA may define or support protocols for agentic discourse beyond simple request-response. This includes supporting standards like the Contract Net Protocol for bidding, auction protocols for dynamic pricing, and protocols for consensus-building.
- **Open APIs & Components - Economy-Enabled Components:** A new specification for "Economy-Enabled ODA Components" could be explored, where a component might not only just declare its technical API but also economic terms (e.g., pricing models) and a digital contract template, potentially making it a proactive offer in an agent marketplace.

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5.5. Architecting for Machine-Native Value Exchange

To explore realizing an Economic Framework, ODA's core business support systems need to evolve beyond human-centric, batch-oriented models, they might be re-architected to accommodate high-velocity, micro-transactions to support autonomous agents negotiating, consuming, and settling value in real time.

- **Deployment & Runtime - Micropayment Transaction Fabric:** The current ODA Canvas provides a runtime for component lifecycle management but lacks a dedicated layer for secure, atomic value settlement. It should be extended with a **"Micropayment Transaction Fabric"**—a distributed ledger or similar high-integrity system integrated into the Canvas. This fabric would natively enable secure, verifiable, and high-frequency low-value payments directly between software agents and ODA components, forming the settlement backbone for a machine-to-machine economy.
- **Billing & Revenue Management - Micropayment-Ready Engine:** Existing Billing components are designed for periodic invoicing or subscription management, not for continuous, granular value flows. The Billing component might evolve to handle streaming micropayments and real-time revenue sharing. It might be able to handle high-throughput usage events, enforce pre- and post-paid spending limits in milliseconds, and to manage instantaneous revenue sharing and profit distribution across complex, dynamically formed collaboration chains between agents, services, and resource providers
- **Open APIs - Micro-transaction API:** Current Open APIs are structured for business transactions initiated by or on behalf of human users. A new **"High-Velocity Transaction API"** specification could be explored for high-volume, machine-initiated micro-transactions. This API could also integrate with digital wallet infrastructures and smart contract platforms, and align with the event-driven principles already established in ODA to enable seamless interoperability for machine-native economic exchanges.

5.6. Assuring the Agent Ecosystem: From QoE to Agentic Performance

Finally, to manage this new complexity, ODA's assurance concepts might extend to the agents themselves.

- **ODA Governance - Experience Assurance for Agents:** Extend ODA's Experience Assurance concepts from human-centric QoE to agent-centric performance. Develop agent-specific KPIs and observability frameworks that track collaboration success metrics, negotiation outcomes, and the overall health of the agent ecosystem.

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By evolving across these six dimensions, ODA will transform from a static blueprint into the dynamic, trusted orchestration platform for the global, autonomous, and economic Internet of Agents. It explores providing a standardized environment where the foundational requirements of the IoA could be met. This evolution could transform a business from a static set of IT systems into a dynamic internal marketplace of capabilities.

ODA Layer	Required Evolution for IoA & Agentic Collaboration	Rationale & Expected New Specifications
ODA Governance	Could evolve from project/organization-level governance to include runtime agent governance. This requires mechanisms for dynamic policy enforcement, ethical boundary management, trust & reputation scoring, and automated compliance for micro-transactions.	**Dynamic & Runtime Governance:** Explore frameworks for governing autonomous agent behaviour. This could include registries for agent identities and verifiable credentials, audit mechanisms for agent decisions, and policy engines that can enforce rules in real-time during agent negotiations.
Business Architecture	The Business Capability Model and Value Streams could be explicitly linked to agent-exposable services. Business capabilities might need machine-readable semantic definitions to become discoverable in an agentic marketplace.	**Value Network Modeling & Dynamic Capability Marketplace:** Explore evolving the static capability map into a model of dynamic **Value Networks**. Define formal, machine-readable semantics for capabilities (using ontologies) so they can be published. Enable the modeling of new value-exchange mechanisms, such as micro-transactions and revenue-sharing models that agents can negotiate.
Information Systems Architecture	The Information Framework (SID) could serve as the core for developing shared domain ontologies. The Data Mesh concept might evolve to support secure, agent-accessible data products with verifiable provenance.	**Ontology-Driven Semantics & Data Mesh:** Formalize the SID into a machine-reasoning ontology to provide the "common language" for agents. Embed the **Data Mesh** principles as the standard data architecture, enabling agents to discover and access consistent data.
Open APIs & Components	Open APIs might benefit from semantic annotations for agent understanding (purpose, non-functional properties). ODA Components/PBCs could expose an "Agent Interface" with economic terms and a digital identity.	**Agent-Native Interfaces & Economy-Enabled Components:** Explore extending Open APIs with semantic annotations for machine-readable discovery of non-functional properties (cost, latency, geo-restrictions). Define an **"Agent-Ready Component"** specification where each component exposes its capabilities, economic

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ODA Layer	Required Evolution for IoA & Agentic Collaboration	Rationale & Expected New Specifications
		terms, and a digital contract template for autonomous negotiation.
Deployment & Runtime	The ODA Canvas could function as a trusted execution environment and agent runtime platform. It might need to host dynamic agent registries, matchmaking engines, and provide secure sandboxes.	**Trusted Agent Runtime & Micro-transaction Fabric:** Explore enhancing the ODA Canvas to support secure deployment and sandboxing of autonomous agents. Integrate a **"Transaction Fabric"** to enable micro-payments between agents. Provide runtime services for agent discovery, reputation management, and secure communication.

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6. Agentic Collaboration Framework

The Agentic Collaboration outline the components required for effective collaboration between autonomous agents based on ontologies.

Figure 3: Agentic Collaboration

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6.1. Intent & Value

This domain represents the high-level business objectives and value streams that the agentic ecosystem is designed to support.

- **Business Objectives:** Defines the overall goals and priorities of the organization. These objectives provide the context for agent behaviour and ensure that agents are working towards the same goals.
- **SLAs (Service Level Agreements):** Specifies the performance and availability requirements for services provided by agents. These SLAs ensure that services are delivered at the required level of quality and reliability.
- **Value Streams:** Represents the sequence of activities that create value for customers. These value streams provide a framework for understanding how agents contribute to the overall business value.

This domain ensures that the agentic ecosystem is aligned with the business goals and that it is delivering value to customers. It provides the context for agent behaviour and ensures that agents are working towards the same objectives.

6.2. Orchestration & Governance

This domain provides the mechanisms for managing and governing the agentic ecosystem, ensuring that agents operate in a coordinated and trustworthy manner.

- **Marketplace:** Provides a platform for agents to discover and acquire services from other agents. This allows agents to easily find the resources they need to perform their tasks and to dynamically adapt to changing conditions.
- **Discovery:** Enables agents to locate and identify other agents and resources within the ecosystem. This allows agents to dynamically discover new opportunities for collaboration and to adapt to changing conditions.
- **Policy:** Defines the rules and regulations that govern agent behaviour and interactions. These policies ensure that agents act in a consistent and predictable manner, and that their actions are aligned with the overall goals of the ecosystem.
- **Trust:** Establishes mechanisms for assessing and managing the trustworthiness of agents. This ensures that agents can rely on each other to act honestly and reliably, and that the ecosystem is protected from malicious actors.

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This domain is essential for ensuring the stability, security, and efficiency of the agentic ecosystem. It provides the mechanisms for managing agent interactions, enforcing policies, and building trust.

6.3. Agent Interconnection

This domain comprises the autonomous agents that perform specific tasks and interact with each other to achieve common goals.

- **Business Agents:** Represent business entities or functions, such as customer relationship management, order processing, or billing. They are responsible for executing business processes and interacting with other agents to fulfil customer needs.
- **Service Agents:** Provide specific services or functionalities, such as network management, security monitoring, or data analytics. They are responsible for delivering these services to other agents and ensuring their quality and reliability.
- **Resource Agents:** Manage and control physical or virtual resources, such as network devices, servers, or storage systems. They are responsible for allocating resources, monitoring their performance, and ensuring their availability.

This domain is the heart of the agentic ecosystem, where the actual work is performed and value is created. The agents in this layer collaborate and coordinate their actions to achieve complex goals.

6.4. Ontology & Knowledge

This domain provides the shared knowledge and understanding necessary for agents to communicate and reason effectively.

- **Shared Vocabularies:** Defines a common set of terms, concepts, and relationships that agents can use to exchange information and understand each other's messages. This ensures that agents are speaking the same language and can interpret each other's intentions correctly.
- **Rules:** Specifies the constraints, policies, and guidelines that govern agent behaviour and interactions. These rules ensure that agents act in a consistent and predictable manner, and that their actions are aligned with the overall goals of the ecosystem.
- **SID Alignment:** Aligns the ontology and knowledge layer with the TM Forum's Shared Information/Data (SID) model. This ensures that the agentic ecosystem is compatible with existing industry standards and best practices, and that it can interoperate with other systems that use the SID model.

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This domain is critical for enabling semantic interoperability between agents, allowing them to understand each other's intentions and coordinate their actions effectively.

6.5. Capability Exposure

This domain focuses on exposing the capabilities of resources and creating digital representations of physical entities, enabling agent and system to understand and interact with the real world.

- **Capability Exposure:** Involves making the functionalities and services of resources accessible to agents through standardized interfaces and protocols. This allows agents to discover and utilize the capabilities of other resources without needing to understand their internal implementation details. APIs (Application Programming Interfaces) are a common mechanism for capability exposure.

This domain bridges the gap between the physical and digital worlds, allowing reasoning about and interact with real-world entities through their digital representations.

6.6. Physical/Virtualized Resources

This domain represents the physical and virtualized infrastructure upon which the entire agentic ecosystem operates. It encompasses the essential resources necessary for agents to function, communicate, and access data.

- **Digital Twin:** Creates virtual representations of physical assets, processes, and systems. These digital twins provide real-time view of the physical world, enabling systems and agents to monitor, analyze, and control physical entities.
- **Network:** Provides the communication pathways for agents to interact, exchange information, and coordinate actions. This includes both physical networks (e.g., Ethernet, Wi-Fi) and virtualized networks (e.g., software-defined networks, virtual private networks). The network infrastructure will require to be reliable, secure, and capable of handling the communication demands of the agent population.
- **Compute:** Offers the processing power required for agents to execute their tasks, perform computations, and make decisions. This can include physical servers, virtual machines, cloud computing resources, and edge computing devices. The compute infrastructure should be scalable and adaptable to accommodate the varying computational needs of different agents.
- **Data:** Provides the storage and access mechanisms for agents to retrieve, process, and persist data. This includes databases, data lakes, data

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warehouses, and other data storage solutions. The data infrastructure will be required to ensure data integrity, security, and accessibility for authorized agents.

This domain is crucial because it provides the basic building blocks for the entire agentic ecosystem. Without a robust and reliable infrastructure, agents cannot effectively operate or collaborate.

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7. Alignment with Work in Digital Ecosystem

Strategically integrating with the ecosystem reference of GB1067, the automation partnering capabilities of GB1027, ODA can become the foundational architecture for the emerging "Internet of Agents" in telecommunications and beyond. The Agentic Collaboration Framework represents in this document a natural evolution of TM Forum's ecosystem vision, extending ODA from a component-based architecture for human-designed systems to a dynamic platform for autonomous AI collaboration.

GB1067 Ecosystem Reference Architecture provides the structural and governance framework for the entire ecosystem. It is divided into planes that directly support and constrain the agentic world:

- **Governance Plane:** This is critical for agents. It provides the rules of the game—policies, participant lifecycle management, and enforcement mechanisms. In an agentic world, these rules are not just static documents; they are machine-readable policies that agents can query and adhere to. This plane ensures that autonomous agents operate within legal and commercial boundaries.
- **Execution Plane:** This provides the composable platform and intelligence capabilities. It is where agents would "live" and execute their tasks. The "Experience Assurance" functional area aligns with the agentic goal of proactive, white-glove customer service .
- **Value Plane:** This is the economic engine. It provides the marketplace and asset management capabilities. For agents to trade, they need a place to list offerings (Marketplace) and a way to manage the digital assets they are trading (Assets). This plane enables the monetization of agent-driven services.

GB1027 Zero-Touch Partnering Reference Architecture provides the specific processes for how partners (or their agents) onboard, interact, and settle accounts. It is the operationalization of the partnership lifecycle.

- **Partner Management & Onboarding:** When an agent from a new service provider wants to join the ecosystem, the processes defined in GB1027 (Partner Portal, Partner Relationship Management) are executed, ideally autonomously by agents governed by the GB1067 Governance Plane.
- **Quote to Order & Settlement:** This is where agent negotiations turn into commercial reality. An agent might negotiate a deal (using the Agentic Framework's intent-based negotiation), which then triggers an automated "Quote to Order" orchestration (as defined in GB1027). Finally, the "Rating, billing and settlement" processes enable the agents to autonomously

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manage the financial aspects of their collaboration, feeding into the GB1067 Value Plane.

IG 1486 The Agentic Collaboration Framework is the top-most, cognitive layer. It envisions a society of autonomous agents with **dynamic behaviour**: discovery, negotiation, and collaboration. It relies on semantic understanding enabling agents to work together even if they were not explicitly programmed to do so.

Phase	Focus Area	Key Actions
Phase 1: Semantic Foundation	Ontology Integration	• Adopt TMF Intent Ontology (TIO) as baseline for agent communication • Extend ODA Component Directory with agent capability schemas • Define IRPs for agent-to-ODA component interactions
Phase 2: Autonomous Orchestration	Zero-Touch Agent Onboard	• Map GB1027 ZTP APIs to agent negotiation protocols • Implement agent discovery registry within ODA Canvas • Develop agent-aware experience assurance metrics
Phase 3: Dynamic Value Exchange	Agentic Marketplace	• Integrate micro-transaction settlement into ODA billing components • Enable value network modeling (GB1067) for agent collaboration scenarios • Launch pilot agent marketplace with trusted partner ecosystem
Phase 4: Runtime Governance	Agentic Control Plane	• Deploy policy-as-code engine for autonomous decision oversight • Implement explainability and audit trails for agent actions • Establish cross-framework governance working group

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8. Scenarios and Use Cases

8.1. Scenarios: Value-driven Agentic AI Marketplace for Iconic Event Management

This scenario presented in IG1502 demonstrates an Agentic AI Marketplace that autonomously orchestrates multi-vendor connectivity and value-added services (e.g., 5G slices, neutral-host DAS, edge compute, security analytics) for large-scale, multi-venue sporting events like the Olympics. The Event Organizing Committee and Venue Operators submit outcome-based business intents (e.g., "support 60,000 attendees with security management") rather than technical specifications.

The marketplace's core operation is founded on the collaboration of a suite of specialized, autonomous agents that perform distinct roles to fulfil the intent:

- A **Capability Discovery Agent** queries participating CSPs and partners via TM Forum Open APIs to discover available network and service capabilities.
- A **Recommendation Agent** interprets the intent and considers policies, SLAs, and commercial rules.
- A **Service Composer Agent** assembles a compliant, multi-vendor service solution from the discovered capabilities, validating policy and SLA alignment.
- An **Orchestration Agent** activates the composed solution across all provider domains using TM Forum Open APIs (e.g., TMF622, TMF641).
- During operations, a **Telemetry Agent** monitors performance, and an **Adaptation Agent** autonomously optimizes services (e.g., expanding 5G slice bandwidth) in response to demand surges to maintain SLAs.

This collaborative agent network enables dynamic, intent-driven service composition, provisioning, and assurance, transforming the CSP from a connectivity provider into an ecosystem orchestrator.

8.2. Scenarios: AI-native Telco Ecosystem for MVNO Vertical Market Expansion

This scenario presented in IG1503 demonstrates an AI-native Telco Ecosystem where a CSP exposes its AI and automation capabilities as monetizable platform services. It enables a Mobile Virtual Network Operator (MVNO) to evolve from a connectivity reseller into a Vertical Digital Service Aggregator, co-creating tailored digital offerings with Small and Medium Business (SMB) vertical partners (e.g., a retail chain) for specific customer segments like elderly shoppers.

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Agentic collaboration is central to realizing two key, AI-native ecosystem processes:

1. **Zero-Touch Partner Onboarding:** The MVNO and a vertical SMB express their partnership **business intent** (covering business, financial, contractual, and operational models) via an AI Copilot. This intent is decomposed, and specialized **AI agents** are instantiated to handle each expectation domain in parallel (e.g., a financial analysis agent). A **master coordinating agent** orchestrates these specialized agents, interacts with ODA components for data, and finally works with the Agreement Management component to generate a formal Agreement Specification—all with minimal human intervention.
2. **Product Offering Co-Creation:** The MVNO and vertical SMB use an **AI Copilot** and collaborative workspace to co-design a product (e.g., an "ElderCare Digital Companion" bundle). An **Offering Co-Creation AI capability** (an applied AI service) orchestrates the process. It leverages **AI agents** and platform intelligence to recommend reusable components, validate compatibility with the catalog, and ensure governance compliance, translating collaborative business intent into a market-ready offering.

In this model, the CSP's platform provides the foundational **Agentic AI Services** and **Intent Management** that power this cross-organizational, agent-driven collaboration.

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9. Terms & Abbreviations Used within this Document

9.1. Abbreviations & Acronyms

Abbreviation / Acronym	Abbreviation/Acronym Spelled Out
AI	Artificial Intelligence
AN	Autonomous Networks
API	Application Programming Interface
BPMN	Business Process Model and Notation
CAPEX	Capital Expenditure
CSP	Communication Service Provider
CTO/CIO	Chief Technology Officer / Chief Information Officer
DLT	Distributed Ledger Technology
DID	Decentralized Identifier
GPU	Graphics Processing Unit
IoA	Internet of Agents
IoT	Internet of Things
JSON-LD	JSON for Linking Data
KPI	Key Performance Indicator
ML	Machine Learning
ODA	Open Digital Architecture
OPEX	Operational Expenditure
OWL	Web Ontology Language
PBC	Packaged Business Capability
QoE	Quality of Experience
RDF	Resource Description Framework
SID	Shared Information/Data Model
SLA	Service Level Agreement
TIO	TM Forum Intent Ontology
VC	Verifiable Credential
ZTP	Zero-Touch Partnering

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10. References

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[3] The Internet of Agents (IoA): A Paradigm Shift in AI Collaboration, *The Internet of Agents (IoA): A Paradigm Shift in AI Collaboration*

[4] Powering the future of telecom: Microsoft brings agentic AI to life at TM Forum DTW, *Powering the future of telecom: Microsoft brings agentic AI to life at TM Forum DTW - Microsoft Industry Blogs*

TMF Assets:

1. GB1067 Ecosystem Reference Architecture
2. GB1027 Zero-Touch Partnering Reference Architecture
3. IG1245 Principles to Define ODA Components
4. IG1464 Value Ecosystem Modelling in the Age of Agents
5. IG1502 Value driven Agentic AI Marketplace for iconic event management Use Case: Reference Implementation
6. IG1503 AI-native Telco Ecosystem for MVNO Vertical Market Expansion
7. IG1454 MCP-T Framework for agentic evolution of AN
8. IG1454A MCP-T Box of Tools for Agentic Evolution of AN
9. IG1454B MCP-T Trusted capability marketplace framework for agentic evolution of AN
10. TMF Open Digital Architecture (ODA)
11. TMF792 Collaboration Model Management
12. TMF793 Collaboration Management

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11. Administrative Appendix

11.1. Document History

11.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	27 Mar 2026	Yun LI	Initial document.

11.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	27 Mar 2026	Yun LI	Initial document.

11.2. Acknowledgments

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